4176 Proposal – Milestone 1

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Anchored: Civic Utility App for Halifax

App Concept and Problem Statement

Anchored is a cross-platform mobile app that helps Halifax residents manage municipal services and stay informed about local civic issues. It centralizes information such as garbage collection schedules, street closures, and reported civic issues into a single, personalized interface.

Municipal service information in Halifax is currently fragmented across open data portals and the 311 system, requiring residents to manually search for updates. The lack of personalization and reminders often leads to missed services and reduced awareness of nearby civic issues.

Target Audience

Halifax residents that are responsible for municipal services or need to be aware of civic issues in their area.

User Needs

- Reliable reminders and calendar integration for waste collection
- Awareness of nearby civic issues
- An easy way to track issues they have reported or saved

Core Features and Functionalities

Authentication Module

- Anchored requires users to authenticate using email and password
- Authentication will be implemented using Supabase.

API Integration

- Halifax Open Data (Waste collection schedules and service zones, street closures and construction data, public civic issue datasets)

Data Storage

- Simple Data (Local Storage)
    - User interface preferences
    - Notification settings
    - Temporarily saved photo
- Complex Data (Database Storage)
    - User Credentials
    - User profile information & Address
    - Saved service reminders

Native Features:

- Camera access for capturing photos of civic issues
- Push notifications for garbage collection reminders and issue updates
- Device location services for filtering nearby issues

Maps Integration

- Google Maps on Android, Apple Maps on iOS
- Displaying the user's live location
- Showing civic issues as custom markers
- Allowing users to select markers to view issue details
- Display issues to users based on radius (distance to user)

Key Technical Challenges

- Handling frequently updated civic data
- Location-based filtering accuracy
- Cross-platform differences in map behavior
- Scheduling notifications based on time-based data

App Architecture

High-Level Structure

UI Components: Reusable components such as cards, lists, and map markers for displaying reminders and civic issues.

Navigation Layer:

- Authentication (login/register)
- Home (dashboard for viewed civic issues and upcoming service reminders)
- Map (civic issues in my area, redirect to civic issue reporting site)
- Settings (address, notifications, permissions, maybe theme)

Data Layer:

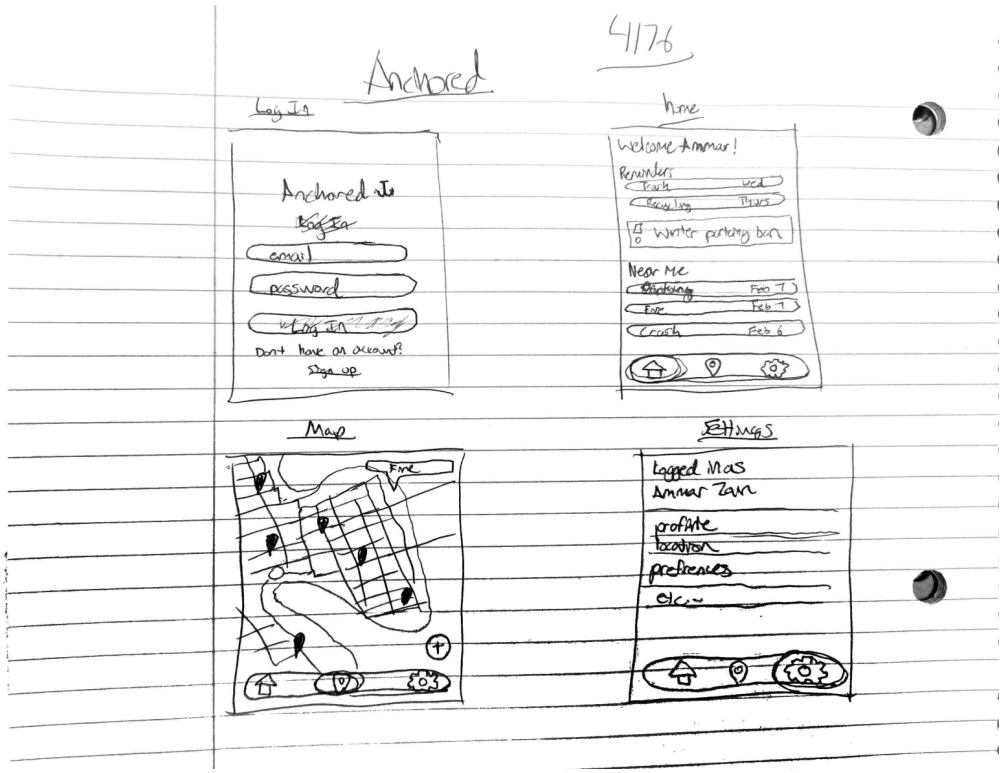
- API service layer
- Local caching
- Database storage

Mockups and User Interaction Flows

Planned Screens:

- Login screen
- registration screen
- Home dashboard showing upcoming service reminders
- Map screen displaying civic issues
- Issue details screen
- User profile and settings screen

Mockups



Key Interaction Flows

Authentication & Setup Flow:
Authenticate → Navigate to Settings → Enter or update address → Load personalized service and issue data

Reminder Flow:
View service schedule → Enable notifications → Receive scheduled reminders

Map Flow:
View map → Select issue marker → View issue details

Issue Reporting (Photo) Flow:
Tap “+” on map → Prompt to take photo → Capture photo → Save temporarily → Redirect to 311 website.

Development Timeline

Week 1 (Milestone 1 – Due Feb 9):

- Finalize app concept and problem statement
- Define target audience
- Identify required APIs and native features
- Design app architecture and navigation flow
- Create initial mockups (login, home, map, settings)
- Write and submit Project Milestone 1 proposal

Week 2 (Feb 10 – Feb 16):

- Set up React Native project and development environment
- Implement basic navigation
- Authentication functionality

Week 3 (Feb 17 – Feb 23):

- Integrate Halifax Open Data API
- Implement data fetching, filtering, and display
- Store simple user preferences locally

Week 4 (Feb 24 – Mar 2):

- Implement garbage and service reminder logic
- Add notification scheduling
- Database integration

Week 5 (Mar 3 – Mar 9) – Milestone 2 Due:

- Integrate maps and live location services
- Add custom markers or routes
- Prepare demo video and progress report
- Submit code, report, and video for Project Milestone 2

Week 6 (Mar 10 – Mar 16):

- Expand native feature usage
- Refine database usage for complex data
- Improve API integration and error handling

Week 7 (Mar 17 – Mar 23):

- UI/UX polishing
- Testing on both Android and iOS
- Bug fixing

Week 8 (Mar 24 – Mar 30):

- Final feature completion
- Performance testing
- Screenshot collection for final report
- Draft final report sections

Week 9 (Mar 31 – Apr 6) – Milestone 3 Due:

- Final testing and deployment build
- Prepare live demo
- Finalize and submit report
- Deliver in-class presentation
- Submit final source code

Feature Priority

The primary focus will be garbage and service reminders with smart calendar and notification integration. Civic Issue awareness will be implemented next, with issue type prioritized by importance, such as road closures and snow removal before lower impact issues like potholes or illegal parking. The idea is to deliver strong and important core features first before moving onto secondary functionality.